

RUHNU, Estonia

WMO: 262270

Lat: 57.7833N

Lon: 23.2589E

Elev: 10

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.4	14.5	3.6	6.6	9.6	8.9	8.7	17.0	31.3	32.6	27.6	30.6	10.4	60	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	76.2	67.8	73.7	66.5	71.5	65.2	69.5	74.5	67.7	72.3	66.0	70.4	7.7	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.2	72.9	65.9	95.4	71.0	64.1	89.7	69.2	33.6	74.6	32.1	72.5	30.8	70.4	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	23.9	21.5	DB	8.5	79.2	7.3	2.3	3.2	80.9	-1.1	82.3	-5.2	83.6	-10.5	85.3	(4)
(5)			WB	7.5	71.2	6.6	2.2	2.7	72.8	-1.2	74.1	-4.9	75.3	-9.7	76.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	29.4	27.9	32.0	39.8	50.5	58.2	64.7	64.7	57.3	47.7	40.8	36.0		
	DBStd	13.97	7.33	7.91	6.08	4.92	6.19	5.02	3.94	3.90	4.24	5.53	5.25	5.96		
	HDD50	3018	638	620	559	307	67	2	0	0	4	110	277	434		
	HDD65	7083	1103	1040	1024	755	449	211	53	53	232	537	727	899		
	CDD50	1508	0	0	0	3	84	249	457	455	222	38	0	0		
	CDD65	96	0	0	0	0	1	8	44	42	1	0	0	0		
	CDH74	178	0	0	0	0	2	15	106	55	0	0	0	0		
	CDH80	7	0	0	0	0	0	0	4	3	0	0	0	0		
(14) Wind	WSAvg	10.4	12.8	11.3	9.3	8.1	7.9	8.5	8.3	9.4	10.2	11.4	13.3	14.3		
(15) Precipitation	PrecAvg	23.1	1.7	1.4	1.2	1.2	1.5	2.2	2.5	2.8	2.0	2.6	2.2	1.8		
	PrecMax	36.5	3.6	2.9	2.4	3.0	3.5	5.0	4.9	8.9	4.7	4.8	3.9	3.9		
	PrecMin	10.7	0.3	0.1	0.3	0.1	0.4	0.5	0.1	0.0	0.2	0.3	0.2	0.7		
	PrecStd	4.8	0.9	0.6	0.6	0.7	0.9	1.1	1.4	1.9	1.1	1.1	1.0	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.7	40.5	46.9	59.9	72.0	76.5	80.2	79.4	69.1	60.5	50.7	47.6		
		MCWB	42.3	38.0	40.3	48.3	61.9	65.3	70.0	70.6	63.4	56.7	48.9	46.1		
	2%	DB	41.9	38.4	42.8	54.0	67.2	72.3	77.0	75.5	66.8	58.2	49.2	45.9		
		MCWB	40.5	37.1	39.0	45.5	58.5	64.7	68.2	68.7	61.8	55.3	47.5	44.2		
	5%	DB	40.2	37.1	40.4	50.2	63.5	69.3	74.4	73.5	65.2	56.5	48.2	44.3		
		MCWB	38.9	36.0	37.6	43.9	56.4	62.6	66.8	67.1	60.6	53.9	46.7	42.9		
	10%	DB	38.2	35.9	38.5	47.1	60.1	66.8	71.8	71.3	63.5	54.9	47.1	42.7		
		MCWB	37.1	35.0	36.5	42.2	54.6	60.8	65.4	65.4	59.3	52.6	45.7	41.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.5	38.7	41.4	49.4	63.4	68.1	72.1	72.4	64.4	57.8	49.3	46.6		
		MCDB	43.2	39.5	45.7	56.9	69.6	74.1	78.5	77.0	67.6	59.5	50.1	47.3		
	2%	WB	40.9	37.4	39.3	46.7	60.4	65.6	69.8	69.8	62.7	55.9	48.0	44.6		
		MCDB	41.9	38.1	41.8	52.5	65.5	70.7	74.9	74.5	65.9	57.6	48.7	45.6		
	5%	WB	39.0	36.2	38.0	44.4	57.7	63.1	68.0	68.0	61.2	54.3	47.0	43.0		
		MCDB	40.0	36.9	39.9	49.2	62.1	68.3	72.7	72.3	64.3	55.9	47.9	44.1		
	10%	WB	37.2	35.1	36.7	42.6	54.9	61.2	66.3	66.3	59.9	53.0	45.8	41.6		
		MCDB	38.1	35.8	38.2	46.5	59.1	66.1	70.8	70.4	62.9	54.7	46.9	42.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.4	7.5	9.9	11.6	10.8	10.3	9.4	7.8	6.4	4.7	4.4		
		MCDBR	4.4	4.9	9.0	14.9	15.7	14.1	13.3	11.3	9.8	7.0	5.0	4.2		
	5% WB	MCWBR	4.6	4.3	6.3	8.2	8.7	7.7	7.0	6.3	6.7	5.4	4.9	4.3		
		MCWBR	4.3	4.6	8.3	12.9	14.1	12.8	12.0	9.7	8.6	6.2	4.7	3.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.296	0.319	0.342	0.378	0.372	0.372	0.402	0.396	0.364	0.345	0.333	0.293		
		taud	2.198	2.200	2.217	2.287	2.410	2.454	2.390	2.405	2.469	2.412	2.247	2.180		
	Ebn at Noon	Ebn at Noon	172	223	253	263	274	275	263	256	247	216	155	135		
		Edh at Noon	16	25	32	36	34	33	34	31	25	21	16	12		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	110	307	779	1282	1781	1872	1762	1388	891	415	125	72		
	RadStd	RadStd	22	56	98	148	155	108	140	122	108	53	23	9		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page